

Stock Performance Analysis

Linear Regression & Hypothesis Testing with the Fama/French 5 Factor Model

Prepared for:	Fund Management Team
Prepared by:	Finance & Analytics Team — Quantitative Analyst
Date:	June 17, 2026
Universe:	15 US Equities Fama/French 5 Factors
Period:	January 2020 – December 2022 (Daily)

Executive Summary

This report presents the results of a multiple linear regression analysis of 15 US equity stocks held in the fund, using the Fama/French 5 Factor Model as the explanatory framework. One OLS regression model was fitted per stock, with daily excess stock returns as the dependent variable and the five Fama/French factors as independent variables. All regression coefficients were tested for statistical significance at the 5% level. The models were verified to be free of autocorrelation using the Durbin-Watson test.

Finding	Result
Market Risk Premium (Mkt_Prem) significance	100% of stocks — most important factor
Average model R-squared across 15 stocks	> 50% — strong explanatory power
Autocorrelation (Durbin-Watson test)	No autocorrelation detected (DW within [1.6, 2.4])
Most significant factor	Mkt_Prem (Market Risk Premium)
Least significant factor	CMA (Investment — Conservative Minus Aggressive)

Methodology

1. Data Pre-processing

Daily stock prices were converted into daily percentage returns using the formula: $\text{Return}(t) = (\text{Price}(t) / \text{Price}(t-1) - 1) \times 100$. This transformation ensures covariance stationarity of the time series, which is a requirement for valid linear regression. Excess returns were then computed by subtracting the daily risk-free rate (RF) from each stock return, yielding the dependent variable for each regression model.

2. Exploratory Data Analysis (EDA)

EDA was conducted using AAPL as a showcase stock. A pairwise correlation heatmap was generated to identify the strength of linear relationships between the stock's excess returns and each factor. A seaborn pairplot and Implot confirmed a strong positive linear relationship between AAPL excess returns and the Market Risk Premium (Mkt_Prem).

3. Regression Modelling

One OLS (Ordinary Least Squares) regression model was defined and fitted per stock using the Python library Statsmodels (statsmodels.formula.api). The regression formula for each stock is: $\text{Stock_Excess_Return} \sim \text{Mkt_Prem} + \text{SMB} + \text{HML} + \text{RMW} + \text{CMA}$. Regression coefficients (intercept + 5 slopes) and the coefficient of determination (R-squared) were extracted for all 15 models.

4. Hypothesis Testing

All regression coefficients were subjected to two-sided t-tests at a significance level of $\alpha = 5\%$. For each coefficient, if the $p\text{-value} < 0.05$, the factor is deemed statistically significant (True); otherwise it is not significant (False). The null hypothesis H_0 states that the coefficient equals zero (no effect), while the alternative hypothesis H_1 states it differs from zero.

5. Autocorrelation Testing

Regression residuals were visualised and tested for autocorrelation using the Durbin-Watson statistic. A DW value within the range [1.6, 2.4] indicates no meaningful autocorrelation. All 15 models passed this check, confirming the validity of the regression results.

Fama/French 5 Factors

Factor	Name	Description
Mkt_Prem	Market Risk Premium	Excess return of the market portfolio over the risk-free rate ($R_m - R_f$)
SMB	Size	Average return on small stock portfolios minus average return on big stock portfolios
HML	Value	Average return on high book/market portfolios minus low book/market portfolios
RMW	Profitability	Average return on robust profitability portfolios minus weak profitability portfolios
CMA	Investment	Average return on conservative investment portfolios minus aggressive investment portfolios

Results

Table 1: Regression Coefficients (Intercepts and Slopes)

OLS regression coefficients for each of the 15 stocks. The intercept represents the alpha (excess return unexplained by the five factors). Slope coefficients indicate the sensitivity of each stock's excess return to each factor.

Stock	Intercept	Mkt_Prem	SMB	HML	RMW	CMA
AAPL	0.0165	1.3084	-0.1903	-0.7681	0.4220	1.0952
AJG	0.0725	0.8775	-0.2547	0.2357	0.0715	-0.1470
AVB	-0.0554	0.8842	-0.0200	0.4258	0.1476	-0.1122
DLR	-0.0385	0.8551	-0.4794	-0.1684	0.0261	0.2839
ICE	0.0063	0.8324	-0.3174	0.1223	-0.1948	-0.0980
INCY	-0.0116	0.6636	-0.1213	-0.2941	-0.3435	0.4129
KHC	-0.0151	0.7916	-0.3410	0.1873	0.2006	0.7675
LLY	0.1042	0.7415	-0.3375	-0.2807	0.1266	0.8165
MCHP	0.0254	1.4749	0.5156	-0.2847	0.2994	-0.1062
MDT	-0.0660	0.8162	-0.0455	0.3710	-0.0027	-0.1746
MOH	0.0906	0.8920	-0.0457	-0.1674	0.1435	0.6135
NDAQ	0.0561	0.9705	-0.3609	0.0202	-0.1193	0.0601
PRU	-0.0060	1.2878	-0.1153	1.2972	-0.2285	-0.4339
STLD	0.1190	1.1592	0.6608	0.6865	0.2361	-0.0887
TER	0.0285	1.3805	0.5028	-0.4814	0.1204	0.0517

Table 2: Significance of Coefficients (True = significant at 5%) and R-squared

True indicates the coefficient is statistically significant at the 5% significance level (p-value < 0.05). False indicates the factor does not significantly explain the stock's excess return. R-squared measures the overall explanatory power of each model.

Stock	Intercept	Mkt_Prem	SMB	HML	RMW	CMA	R-squared
AAPL	False	True	True	True	True	True	0.8041
AJG	False	True	True	True	False	False	0.5767
AVB	False	True	False	True	False	False	0.4902
DLR	False	True	True	True	False	True	0.3965
ICE	False	True	True	True	True	False	0.5460
INCY	False	True	False	True	True	True	0.2838
KHC	False	True	True	True	True	True	0.4217
LLY	False	True	True	True	False	True	0.2789
MCHP	False	True	True	True	True	False	0.6718
MDT	False	True	False	True	False	False	0.5399
MOH	False	True	False	False	False	True	0.3030
NDAQ	False	True	True	False	False	False	0.6173
PRU	False	True	False	True	True	True	0.8351

STLD	False	True	True	True	False	False	0.5588
TER	False	True	True	True	False	False	0.5998

Factor Importance Ranking

The table below shows the percentage of stocks for which each factor is statistically significant at the 5% level. A higher percentage indicates a more important and broadly applicable factor across the fund's equity universe.

Rank	Factor	% of Stocks Significant	Interpretation
1	Mkt_Prem	100.0%	Dominant driver — significant for all stocks
2	HML	86.7%	Value effect — moderately significant
3	SMB	66.7%	Size effect — broadly significant
4	CMA	46.7%	Investment style — least universal factor
5	RMW	40.0%	Profitability — selective significance

Average Model Quality (R-squared)

The average R-squared across all 15 regression models is **0.5282 (52.8%)**. This indicates that the Fama/French 5 Factor Model explains, on average, more than half of the variation in daily excess stock returns across the fund's equity universe. This is a strong result for a daily returns model and supports the use of factor investing as a framework for understanding equity performance.

Interpretation & Conclusions

The Fama/French 5 Factor Model provides a statistically robust framework for explaining the performance of the fund's 15 equity holdings. The Market Risk Premium is universally significant, confirming that systematic market exposure is the primary driver of return for all stocks in the portfolio. The Size (SMB) and Value (HML) factors provide additional explanatory power for a meaningful subset of stocks, while the Profitability (RMW) and Investment (CMA) factors — introduced in the extended 5-factor model — show more selective significance.

The declining marginal significance of additional factors is consistent with the academic literature: the 3-Factor Model (Mkt_Prem, SMB, HML) captures the bulk of explanatory power, with CMA and RMW offering incremental improvement. The fund management team should consider prioritising market beta management and size/value tilts when constructing or rebalancing the portfolio.

No autocorrelation was detected in any of the 15 regression models, confirming the statistical validity of the OLS estimates and the reliability of the hypothesis testing results presented in this report.